

building, surrounded by reporters and passers-by, pressing a button and making a call, much to the bewilderment of the onlookers and to Joel Engel's chagrin...

What bewildered people in 1973, has absolutely no effect on us now though. These days you would actually turn more heads if you proclaimed, "No, I don't own a cell phone."

All of us seem more interested in buzzwords like 3G, 4G, CDMA, HSCSD, GSM, MMS, Tri-band, GPRS and Bluetooth, because it's what we see mentioned on the feature lists of the phone models we aspire to buy. However, the biggest buzzword of the twenty first century is "convergence", and the most common form of technology has turned out to be the humble cell phone. It doesn't take a genius to figure out why so much innovation has happened in the mobile handset and services markets.

What used to be a device to make and receive calls has now become a manifestation of all the technologies and services we desire. Think about it. When phone calls to foreign countries were prohibitively costly, e-mail and chat rescued the day. However, e-mail is free, but the humble SMS is still the most popular way of greetings and information flow. Every Diwali, Eid or Christmas, we receive more SMSes than e-mails from friends and acquaintances, because it's cheap and instantaneous. It's not as intrusive as a phone call, yet intrusive enough to be delivered instantly—whether you're in the loo, asleep or rock-climbing. Even if your phone is off, you get your messages as soon as it's turned on, which has a much higher probability of occurring than, say, you sitting down at a PC and checking mail. The limitations of SMS (in terms of characters) is now being offset by push-mail services, thanks to our digital data capable networks. So now executives with a BlackBerry are informed of e-mail the way most of the world is informed about a received SMS.

It's not just about messaging systems though, and increases in processing power, advancements in mobile OSes and better batteries mean that you can now buy phones that are, for all practical purposes, mini computers, or PDAs at the least. With the sheer amount of software developers focussing on the mobile segment, and thanks to software platforms like Java, almost every phone can do almost anything. You can view your documents, store your files, click and store pictures, upload those pictures to your site, check mail, surf, chat over IM, watch videos, listen to music or FM, and even play games. What more could you want? Plenty more!

The Future

With 4G promised to all of us soon, we can start seeing PC like bandwidth on mobile phones, which means streaming video content, in the form of subscription services or even dedicated mobile TV chan-

nels! The focus here is entertainment, obviously, so gaming, chatting, streaming radio and TV are what get all the headlines. However, with 4G we're talking about being connected to cell networks at near-LAN speeds (100 Mbps to 1 Gbps). This also brings along with it video conferencing capabilities, true MMS capabilities and mobile peer-to-peer networks.

So will the lines between mobile phones and laptops begin to blur next? Probably, but it might take a decade or two. With the kind of research being done, although notebook computers will get smaller, so will cell phones. Perhaps we might soon see the cell phone that fits into your ear, or one that's embedded in your skin, powered by your heart! The cell phone and notebook computers are different enough to survive un-converged, but we're betting that the differences will not stop manufacturers from trying to converge the two anyway.

Security Concerns

Miniaturisation and mass adoption bring with them security concerns aplenty. Just as PC users have to worry about viruses, Trojans and spyware, the "smart" phone of the future will need to have anti-virus software running on them to keep them safe from hackers. With Wi-Fi access already available at most popular haunts (coffee hops, airports, train stations, etc.) and WiMAX promising to extend networks across entire cities, we're headed towards a time when we will always be connected to data networks, and the Net. The problem is, whatever device you use, public networks are usually teeming with malware, and the odd hacker looking for a chance to get at your personal information.

With miniaturisation we end up with a human problem, where the voyeur hidden amongst us surfaces. Already because of camera phones we're in the age of MMS, mobile video and still photography scandals. Can you imagine what would happen if we had tiny mobiles or cameras built into, say, a pair of sun glasses, and made them cheaply available at every local electronic shop?

The Last Word

The technology trends we've seen, all around us, point specifically to Anywhere Computing. Whether it's just about being connected all the time, or about making smarter devices to allow your fridge to order more milk via your cell phone when you're running low, it's all heading in the same direction. In this magazine itself you have read about grid computing, ubiquitous computing and now anywhere computing, and it all boils down to many more smarter devices, all communicating, all pervasive, and all of them offering you the opportunity to surround yourself in sweet silicon goodness, and call yourself a geek. It's now just a matter of time... the geek really will inherit the Earth! ■

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